



METABOLIC & GLP-1

Semaglutide

Semaglutide – 10 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Semaglutide
SYNONYMS / ALIASES	NN9535; GLP-1 receptor agonist analog
CAS NUMBER	910463-68-2
MOLECULAR FORMULA	C187H291N45O59
MOLECULAR WEIGHT	4113.58 g/mol
AMINO ACID SEQUENCE	HXEGTFTSDVSSYLEGQAAKEFIAWLVRGRG (X = alpha-aminoisobutyric acid / Aib at position 2; Arg at position 34; Lys at position 26 conjugated to a C18 fatty diacid via a gamma-Glu-(2xAEAA) linker)
SOURCE	Synthetic (solid-phase peptide synthesis with side-chain acylation)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	10 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Avoid repeated freeze-thaw cycles. Acylated peptides are prone to aggregation at the air-water interface - handle with minimal agitation.

STRUCTURAL NOTES

Semaglutide is a 31-amino-acid synthetic analog of human glucagon-like peptide-1 (GLP-1). Its backbone derives from native GLP-1(7-37) with an alpha-aminoisobutyric acid substitution at position 2 (conferring resistance to dipeptidyl peptidase-4 cleavage) and an arginine substitution at position 34. A C18 fatty diacid is conjugated to the epsilon-amino group of Lys26 through a gamma-Glu-2x(AEEA) spacer, enabling reversible albumin binding and the extended half-life characteristic of long-acting incretin analogs.

SELECTED PEER-REVIEWED REFERENCES

1. Lau J. et al. - *Discovery of the once-weekly glucagon-like peptide-1 (GLP-1) analogue semaglutide* - Journal of Medicinal Chemistry, 2015 - DOI: 10.1021/acs.jmedchem.5b00726
2. Marso S.P. et al. - *Semaglutide and cardiovascular outcomes in patients with type 2 diabetes* - New England Journal of Medicine, 2016 - DOI: 10.1056/NEJMoa1607141
3. Wilding J.P.H. et al. - *Once-weekly semaglutide in adults with overweight or obesity* - New England Journal of Medicine, 2021 - DOI: 10.1056/NEJMoa2032183

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

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Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.